# ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 2 | **(c)** hypotonic to the cell | Water entered. |
| 3 | **(a)** synthesis of ATP | Mitochondria do ATP. |
| 4 | **(a)** cell wall | Rigid wall. |
| 5 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 6 | **(a)** hypertonic to A and hypotonic to B | Different internal concentrations. |
| 7 | **(c)** bacterial | Prokaryote. |
| 8 | **(b)** large central vacuole | Takes up space. |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | ATP is the energy currency of the cell, which is why mitochondria are called powerhouses. |
| 10 | **(b)** Both A and R are true but R is not the correct explanation of A. | Both are true, but the composition of the wall does not explain why animal cells lack it. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | SA/V = 6 and 2 (1); Larger cell exchanges less efficiently, so it divides (1). | 1+1 |
| 12 | It protects the cell and controls entry and exit of substances (1); It allows only some substances to pass (1). | 1+1 |
| 13 | Protoplasm separates from the cell wall (1); Gap between wall and membrane (1). | 1+1 |
| 14 | 4 cells (1); 46 chromosomes in each (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | 0.5 per cent: gains mass (1); 5 per cent: loses mass (1); 1 per cent is isotonic with the potato cell sap (1). | 1+1+1 |
| 16 | Break cells gently (1); Centrifuge at low speed to remove nuclei (1); Higher speed to get mitochondria (1). | 1+1+1 |
| 17 | Fill the bag with starch and glucose (1); Place in water (1); Glucose passes out; starch stays (1). | 1+1+1 |
|  | **OR:** Root tip: mitosis, growth (1); Testis: meiosis for gametes (1); Chromosome numbers (1). | 1+1+1 |
| 18 | They contain chlorophyll (1); They trap sunlight (1); They make food (glucose) by photosynthesis (1). | 1+1+1 |
|  | **OR:** Ability of a cell to form a whole organism (1); Plant cells are totipotent (1); Used to grow plants from single cells (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Osmosis is the movement of water molecules through a selectively permeable membrane from a region of higher water concentration to a region of lower water concentration (1). (ii) Water enters the raisin cells by osmosis because the surrounding water is hypotonic to the cell contents (1). (iii) The strip in the salt solution (1). (iv) Salt solution: $-10\%$ (loses water); distilled water: $+12\%$ (gains water) (1); the cell loses water in a hypertonic medium and gains water in a hypotonic medium (1). | 1+1+1+2 |
| 20 | Prokaryotic: no true nucleus, no membrane-bound organelles (1); Eukaryotic: nucleus and organelles (1); Eukaryotic cells divide the work among organelles (1); Viruses have no cell structure (1); Viruses multiply only inside host cells (1). | 1+1+1+1+1 |
|  | **OR:** Mammalian red blood cell (1); Biconcave disc, large surface (1); More space for haemoglobin (1); No nucleus means no repair or division (1); Short life of about 120 days (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Saying that bacteria have a nucleus, or that prokaryotes have no genetic material; the DNA is present but is not enclosed in a nuclear membrane.
- Mixing up hypotonic and hypertonic: in a *hypotonic* outside solution the cell swells, in a *hypertonic* one it shrinks.
- Calling osmosis the movement of any solute; osmosis is the movement of *water* through a selectively permeable membrane.
- Confusing the functions of rough ER (protein synthesis, studded with ribosomes) and smooth ER (lipids, detoxification), or saying that mitochondria 'make' energy; they release energy from food as ATP.
- Saying that the cell wall and cell membrane are the same; the membrane is selectively permeable and present in all cells, the wall is rigid and fully permeable.
- Forgetting that plant cells do not burst in water because of the cell wall, whereas an animal cell like RBC can swell and burst.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** energy release | Respiration. |
| 2 | **(c)** vacuole | Water and solutes. |
| 3 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 4 | **(d)** 16 | 2^4. |
| 5 | **(a)** proteins | Protein synthesis. |
| 6 | **(d)** petals and ripe fruits | Pigments. |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 8 | **(a)** 8 | 2 x 2 x 2. |
| 9 | **(c)** A is true but R is false. | Ribosomes are non-membrane-bound particles, so R is false. |
| 10 | **(d)** A is false but R is true. | Prokaryotic cells have no nuclear membrane; their DNA lies free in the cytoplasm, so A is false. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Mitochondria (1); Cellular respiration (1). | 1+1 |
| 12 | Plant cells have a cell wall (1); Animal cells do not (1). | 1+1 |
| 13 | Makes ribosomal subunits (1); Ribosomes make proteins (1). | 1+1 |
| 14 | Enzymes digest the cell (1); Autolysis (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Mitosis: 8 in each cell (1); Meiosis I: 4 chromosomes (each with two strands) in two cells (1); Meiosis II: 4 chromosomes in four cells (1). | 1+1+1 |
| 16 | Tiny channels in the cell wall (1); Connect cytoplasm of adjacent cells (1); Exchange of substances (1). | 1+1+1 |
| 17 | For growth (1); For repair of damaged tissues (1); For replacement of old cells (1). | 1+1+1 |
| | **OR:** Mature cells store water and maintain turgor (1); Meristem cells are dividing and dense (1); Vacuoles enlarge on maturity (1). | 1+1+1 |
| 18 | Volume ratio = (25)^3 = 15625 (1); SA/V is inversely proportional to the diameter (1); Bacterium has 25 times larger SA/V (1). | 1+1+1 |
| | **OR:** Proteins are not packaged (1); Not sent out by vesicles (1); Secretion fails (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Nucleus (1). (ii) Mitochondria (1). (iii) Chloroplast and the large central vacuole (and cell wall) (1). (iv) Rough endoplasmic reticulum (has ribosomes that make proteins) and the Golgi apparatus (1) which modify, package and dispatch them (1). | 1+1+1+2 |
| 20 | Water enters by osmosis (1); Turgor pressure builds up (1); Wall pressure balances (1); Equilibrium: no net movement (1); Protoplast without wall bursts (1). | 1+1+1+1+1 |
| | **OR:** Equal discs washed (1); Water at different temperatures (1); Measure colour intensity or absorbance (1); Greater leakage at higher temperatures (1); Heat damages proteins and lipids of the membrane (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Saying that bacteria have a nucleus, or that prokaryotes have no genetic material; the DNA is present but is not enclosed in a nuclear membrane.
- Mixing up hypotonic and hypertonic: in a *hypotonic* outside solution the cell swells, in a *hypertonic* one it shrinks.

- Calling osmosis the movement of any solute; osmosis is the movement of *water* through a selectively permeable membrane.
- Confusing the functions of rough ER (protein synthesis, studded with ribosomes) and smooth ER (lipids, detoxification), or saying that mitochondria 'make' energy; they release energy from food as ATP.
- Saying that the cell wall and cell membrane are the same; the membrane is selectively permeable and present in all cells, the wall is rigid and fully permeable.
- Forgetting that plant cells do not burst in water because of the cell wall, whereas an animal cell like RBC can swell and burst.

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# ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 2 | **(d)** 2 : 1 | 46 : 23. |
| 3 | **(a)** turgor in cells | Osmosis. |
| 4 | **(b)** leucoplast | Amyloplast. |
| 5 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 6 | **(a)** half | SA/V = 6/a. |
| 7 | **(a)** heat denatures the membrane proteins | Leaky membrane. |
| 8 | **(a)** no membrane-bound organelles | Simple cell. |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | Water enters by osmosis, and the wall prevents bursting. |
| 10 | **(b)** Both A and R are true but R is not the correct explanation of A. | Both are true, but they describe different organelles. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Rapid production of identical plants (1); Disease-free plants (1). | 1+1 |
| 12 | Underground, no light (1); They are growing cells (1). | 1+1 |
| 13 | Osmosis is passive, active transport needs energy (1); Osmosis is along gradient; active is against it (1). | 1+1 |
| 14 | Division of work (1); Increases efficiency by compartments (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Nucleus gives mRNA (1); Ribosomes on rough ER make protein (1); Golgi packs; vesicles go to the membrane (1). | 1+1+1 |
| 16 | Initially isotonic (1); Solute may enter or water may evaporate (1); Concentration changes (1). | 1+1+1 |
| 17 | High solute outside (1); Microbes lose water (1); Cannot multiply (1). | 1+1+1 |
| | **OR:** No cell structure (1); Cannot reproduce on their own (1); Bacteria have ribosomes, membranes, own metabolism (1). | 1+1+1 |
| 18 | Epidermis protects (1); Palisade packed with chloroplasts for light (1); Structure related to function (1). | 1+1+1 |
| | **OR:** Small organism: short distances, high SA/V (1); Large body: long distances (1); Needs circulatory and respiratory systems (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Slide P (1). (ii) Slide Q (1). (iii) Prokaryote: bacterium (e.g. *E. coli*); eukaryote: onion plant, human, yeast, Amoeba (1). (iv) Any two: prokaryotes have no nuclear membrane, eukaryotes have; prokaryotes have no membrane-bound organelles, eukaryotes do; prokaryotes are generally much smaller (1+1). | 1+1+1+2 |
| 20 | Common: nucleus, mitochondria, ER, Golgi, ribosomes, lysosomes (1); Plant only: wall, plastids, large vacuole (1); Animal only: centrioles (1); Structure of each (1); Function of each (1). | 1+1+1+1+1 |
| | **OR:** Protects the cell contents (1); Selective permeability controls entry and exit (1); Receptors on the surface receive signals (1); Keeps concentrations inside stable (1); Hence cell survives in changing surroundings (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

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